THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 3: **The Atomic Foundations of Matter** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The formula unit mass of the compound $Ca_3(PO_4)_2$ is (Ca = 40, P = 31, O = 16): 1

(a) 262 u **(b)** 310 u **(c)** 230 u **(d)** 278 u

2. The mass of hydrogen in 36 g of water is (H = 1, O = 16): 1

(a) 32 g **(b)** 4 g **(c)** 8 g **(d)** 2 g

3. Assertion (A): A molecule of nitrogen has a triple bond. Reason (R): Each nitrogen atom needs three more electrons to complete its octet. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. The mass of 3 formula units of $NaCl$ relative to 1 formula unit of MgO (Mg = 24) is: 1

(a) 175.5 : 40 **(b)** 58.5 : 40 **(c)** 58.5 : 24 **(d)** 117 : 40

5. When 20 g of sulphur burns in oxygen producing 40 g of sulphur dioxide, the mass of oxygen used is: 1

(a) 40 g **(b)** 60 g **(c)** 10 g **(d)** 20 g

6. Assertion (A): Atoms are the smallest particles of matter which cannot be divided any further. Reason (R): Atoms consist of electrons, protons and neutrons. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. What mass of oxygen is present in 100 g of $CaCO_3$ (Ca = 40, C = 12, O = 16)? 1

(a) 16 g **(b)** 12 g **(c)** 32 g **(d)** 48 g

8. 24 g of magnesium burns completely in oxygen to form 40 g of magnesium oxide. The mass of oxygen that combined is: 1

(a) 24 g **(b)** 16 g **(c)** 64 g **(d)** 40 g

9. **Assertion (A):** The chemical formula of aluminium oxide is Al_2O_3. 1
Reason (R): Aluminium has valency 2 and oxygen has valency 3.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Sodium chloride exists as separate $NaCl$ molecules. 1
Reason (R): Ionic compounds consist of oppositely charged ions held together by electrostatic forces.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does an ionic compound dissolve in water but not in kerosene? 2

12. Calculate the molecular mass of $C_{12}H_{22}O_{11}$ (C = 12, H = 1, O = 16). 2

13. Calculate the mass of Na in 100 g of Na_2SO_4 (Na = 23, S = 32, O = 16). 2

14. State why covalent compounds like sugar do not conduct electricity even when dissolved in water. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain the formation of the ionic compound Li_2O with electron transfer and the formula unit mass (Li = 7, O = 16). 3

16. State the postulates of Dalton's theory that explain 3
(a) the law of conservation of mass
(b) the law of constant proportions.

17. Explain why the law of conservation of mass is not violated in the burning of a candle even though it seems to lose mass. 3

OR

Calculate the number of atoms of each kind in 5 formula units of $Al_2(SO_4)_3$ and the total. 3

18. Calculate the masses of calcium and chlorine present in 55.5 g of $CaCl_2$. 3

OR

Compare the charge on the nucleus, the size and the electronic arrangement of Na and Na+. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Two samples of water, one from a well in Pune and one from a river in Kerala, are decomposed. In each, hydrogen and oxygen are found in the mass ratio 1 : 8. A 27 g sample of water is then decomposed in the laboratory. 5
(i) Which law is illustrated by the identical ratio in both samples?
(ii) What fraction of the mass of water is hydrogen?
(iii) Find the mass of hydrogen in the 27 g sample.
(iv) Find the mass of oxygen obtained and say what the ratio would be in a sample made in a factory.

20. A mixture of 20 g of hydrogen and 20 g of oxygen is ignited. Find the mass of water formed, the mass of the excess reactant and the total mass after the reaction. Explain the use of the law of constant proportions. 5

OR

Calcium carbonate decomposes on heating into CaO and CO2. Using atomic masses (Ca = 40, C = 12, O = 16), find the masses of CaO and CO2 obtained from 100 g of CaCO3 and verify conservation of mass. 5

END OF QUESTION PAPER **Total: 40 Marks**